For cold-weather applications where transmission oil SAE 0W-20 is recommended, Cat TDTO Cold Weather is the first choice oil. Second choice oils for cold-weather transmission applications are commercial oils of full synthetic basestock that do not have viscosity index improvers. These oils do meet the Cat TO-4 performance requirements. Typical lubricant viscosity grades are SAE 0W-20, SAE 0W-30, and SAE 5W-30. Oils of lubricant viscosity grade SAE 0W-20, SAE 0W-30, or SAE 5W-30 that contain a Cat TO-4 additive package are a last choice.

The footnotes are a key part of the tables. Read ALL footnotes that pertain to the machine compartment in question.

Table 6

Recommendations for power train, transmissions, and hydraulics Fluids and Cat Performance Requirements		
Preferred	**Minimum Acceptable Performance Requirements for Commercial Oils**	**Application**
Cat HYDO Advanced	Per Product Data Sheet requirements	Hydraulic systems and Hydrostatic Transmissions
Cat Bio HYDO Advanced	Per Product Data Sheet requirement	Hydraulic systems and Hydrostatic transmissions requiring biodegradable fluids
Cat TDTO	Cat TO-4	Power shift transmissions, some direct drive transmissions, final drives, differentials, and transfer drives of many machines
Cat TDTO-TMS	Cat TO-4M	
Cat FDAO, Cat FDAO SYN	Cat FD-1	Highly loaded final drives and axles that do not have friction materials
Cat GO	API GL-5	Gears and roller bearings where extreme pressure additives are required.

Organization of the Tables

Table 9 is for all Cat machine and Expanded Mining products engines. This table applies to all oils that are recommended for the engine crankcase.

Table 8 is for all Cat machine hydraulic systems and machine hydrostatic transmissions. This table applies to all oils that are recommended for the machine hydraulic systems, pumps, and valves. This table applies to all oils that are recommended for the machine hydrostatic transmissions.

All other tables are listed in the product line categories. These tables contain information about other machine and Expanded Mining products compartments and the lubrication recommendations for each compartment. Some machine compartments allow the use of more than one oil type. Some machine and Expanded Mining products compartments restrict the use of some oil types. These tables contain any exceptions to tables 9 and 8 . Consult your Cat dealer if additional information is needed about any of the exceptions.

Special applications are included at the end of this section. Special applications include starting engine and variable pitch fan. If you do not find a compartment in the product category, check this table. Consult your Cat dealer if additional information is needed about special applications.

Note: Exceptions to any information in the common tables are listed in the tables in the product line sections.

All Cat Machine and Expanded Mining Products Engines

Refer to the “General Information for Lubricants” article for important lubricant information.

Supplemental heat is recommended for cold-soaked starts below the minimum ambient temperature. The parasitic load and other factors will determine if supplemental heat is required for cold-soaked starts that are above the minimum temperature that is stated. Cold-soaked starts occur when the engine has not been operated for a time. The oil becomes more viscous due to cooler ambient temperatures.

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For oil recommendations for Tier 4 EPA certified engines, EU stage IIIB and IV type approved engines, and Japan Step IV approved engines refer to the “Recommendation for Tier 4 Engines” section, the “Cat Engine Lubricants Recommendations/ Requirements” table, and the “TBN recommendations for applications in Cat engines” table in Special Publication, SEBU6250, “Engine Oil”.

Refer to the “Lubricant Information” section in this Special Publication for a list of all Cat engine oils.

Cat ECF refers to Engine Crankcase Fluid specifications. Refer to the “Maintenance” section of this Special Publication, “Lubricant Information”, and to the “Cat Engine Crankcase Fluids (ECF) Definitions” table in this Special Publication, “Engine Oil” for details. Commercial alternative diesel engine oils must meet one or more of these Cat ECF specifications. Refer to the “Cat Engine Lubricants Recommendations/Requirements” table in Special Publication, SEBU6250, “Engine Oil”.

The minimum requirements for commercial diesel engine oils are provided in Special Publication, SEBU6250, “Engine Oil (Cat Machine Diesel Engines)”. These requirements apply to all the fluids recommendations given in this Special Publication. Commercial alternative diesel engine oils are, as a group, second choice oils.

Note: Cat Special Application Engine Oil (SAEO), API CF oil of SAE 30 and SAE 40 viscosity grades is recommended for use in 3116 and 3126 marines engines.

Note: SAE 10W-30 is the preferred viscosity grade for the 3116, 3126, C7, C-9, and C9 diesel engines when the ambient temperature is between -18° C (0° F) and 40° C (104° F).

Note: C175 Series diesel engines require the use of **multigrade** SAE 40 oil. For example: SAE 0W-40, SAE 5W-40, SAE 10W-40, or SAE 15W-40. In ambient temperatures of −9.5° C (15° F) or above, SAE 15W-40 is the preferred oil viscosity grade

If ambient temperature conditions at engine start-up require the use of multigrade SAE 0W oil, SAE 0W-40 viscosity grade is preferred over SAE 0W-30.

Table 7

Cat Engine Lubricants and Viscosities for Ambient Temperatures(1)							
Compartment or System	Oil Type, name and Performance Requirements		Viscosity Grade	°C		°F	
				Min	Max	Min	Max
Recommendations for all Cat diesel engines	Cat Diesel Engine Oil-Ultra Low Sulfur (API CK-4)(1)	Cat DEO-ULS	SAE 15W-40	-10	50	14	122
			SAE 10W-30	-18	40	0	104
		Cat DEO-ULS SYN	SAE 5W-40	-30	50	-22	122

(continued)

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(Table 7, contd)

		Cat DEO Cold Weather	SAE 0W-40	-40	40	-40	104
	Cat Diesel Engine Oil (API CI-4/ CI-4 PLUS and API CH-4)[1]	Cat DEO[2]	SAE 15W-40	-10	50	14	122
			SAE 10W-30	-18	40	0	104
Specialty oils for specific Cat diesel engines	Cat Multi-Application Engine Oil (Non-Cat mixed fleets)	Cat MAEO Plus	SAE 10W-40	−18	50	0	122
	Cat Precombustion Chamber Oil (Specific applications only)	Cat PCO	SAE 30	-5	40	0	104

(1) Cat engine oils exceed the performance requirements of the respective API categories
(2) DEO oils are formulated for pre-Tier 4 machines, and should not be used for engines with aftertreatment devices.

All Cat Machine Hydraulic Systems and Machine Hydrostatic Transmissions

Refer to the “General Information for Lubricants” article for important lubricant information.

Cat HYDO Advanced 10 SAE 10W, Cat HYDO Advanced 20 SAE 20W, Cat HYDO Advanced 30 SAE 30W, or Cat BIO HYDO Advanced are the preferred oils for use in most Cat machine hydraulic and hydrostatic transmission systems. **Cat HYDO Advanced fluids have at least a 50% increase in the standard oil drain interval** for machine hydraulic systems over second and third choice oils - when following the maintenance interval schedule for oil filter changes and for oil sampling that is stated in the Operation and Maintenance Manual for your particular machine. 6000 hour or higher oil drain intervals are possible when using S·O·S Services oil analysis. Consult your Cat dealer for details. To gain the most benefit from the improved performance designed into Cat HYDO Advanced fluids, when switching to Cat HYDO Advanced fluids, cross contamination with the previous oil should be kept to less than 10%.

Second choice oils are Cat MTO, Cat DEO, Cat DEO-ULS, Cat TDTO, Cat Cold Weather TDTO, Cat TDTO-TMS, Cat Cold Weather DEO-ULS.

Third choice oils are commercial oils that meet Cat ECF-1-a, Cat ECF-2, Cat ECF-3 (which can include viscosity grades not given in Table8), Cat TO-4, or the Cat TO-4M performance requirements, and that have a minimum zinc additive level of 0.09 percent (900 ppm). Commercial biodegradable hydraulic oil must meet the Cat BF-2 specification. Refer to the machine Operation and Maintenance Manual and/or consult your local Cat dealer before using commercial oils that meet Cat BF-2 in Cat Hydraulic Excavators.

The minimum viscosity for commercial alternative oils used in most Cat machine hydraulic and hydrostatic transmission systems is 6.6 cSt at 100 °C (212 °F) (“ASTM D445”).

Note: Industrial hydraulic oils are not recommended for service fill in Caterpillar hydraulic systems.

Note: For machines equipped with hydraulic hammers, do not use viscosity grades SAE 0W or SAE 5W oils. Refer to the “Special Applications” section in this article.

For applications requiring the use of Fire-Resistant hydraulic fluids or Food-Grade hydraulic fluids, refer to the information on these two types of fluids given in the Hydraulic Oil Section of this Special publication.

Commercial oils of viscosity grades that are not included in Table 10 may be used if the oils are per Cat ECF specifications. Refer to the "Cat Engine Crankcase Fluids (ECF) Definitions " table in the "Engine Oil" section of this Special Publication. Commercial oils are second choice.

If noise is a problem in the hydraulic system, 1U-9891 oil additive may be used in the hydraulic system. This additive is a friction modifier that helps reduce the noise level.

Table 8 describes the hydraulic system recommendations for all Cat machines. Some machines require different temperature ranges or specific hydraulic fluids. The recommendations for those machines are given in the respective machine Lubricant Viscosity Table.

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Table 8

Hydraulic Systems Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Hydraulic System and **Hydrostatic Transmissions**	Cat HYDO Advanced 10(1) Cat TDTO	SAE 10W	−20	40	−4	104
	Cat HYDO Advanced 20 Cat TDTO	SAE 20W	−5	45	23	113
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	0	50	32	122
	Cat BIO HYDO Advanced(2)	“ISO 46” Multi-Grade	−30	45	−22	113
	Cat MTO Cat DEO-ULS Cat DEO	SAE 10W-30	−20	40	−4	104
	Cat DEO-ULS Cat DEO	SAE 15W-40	−10	50	5	122
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
	Cat DEO-ULS SYN	SAE 5W-40	-20	50	-22	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104

(1) Typical factory fill oil for standard configuration machines.
(2) Factory fill oil for applications that require biodegradable oils.

Articulated Trucks

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 725 through 740

For the Underground Mining articulated trucks, refer to the Underground Mining section is this article.

For differentials and final drives of Series E II and Series 700, Cat TDTO SAE 50 is preferred in most applications, particularly in continuous operation. If the ambient temperature is below −15° C (5° F), warm up the oil prior to operation by performing the procedures in the Operation and Maintenance Manual, “Engine and Machine Warm-up”. Failure to warm up the oil prior to operation can damage the machine.

For Differentials and Final Drives of Series E II and Series 700, do NOT use SAE 0W-20 oils when the typical daily maximum ambient temperature is above −10° C (14° F).

For hydraulic systems that are not given in Table 9 , refer to the hydraulic system recommendations given in Table 8 .

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Table 9

Articulated Trucks Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Power shift transmissions	Cat TDTO Cold Weather	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	−10	50	14	122
Hoist (Ejector), Steering and Brake System, Brake Control Valve Damper, and Suspension Hydraulic System for E Series II and 700 Series	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104
	Cat TDTO	SAE 10W(1) (2)	−20	50	−4	122
Differentials and Final Drives (except for E Series II Articulated Trucks, and 700 Series Articulated Trucks)	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90 (1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Differentials and Final Drives for E Series II Articulated Trucks and 700 Series Articulated Trucks	Cat TDTO Cold Weather	SAE 0W-20	−40	−10	−40	14
	Cat TDTO	SAE 30	−20	15	-4	59
		SAE 50(1)	−15	40	5	104
	Cat TDTO-TMS	Multi-Grade	−25	22	−13	72
Output Transfer Gear for E Series II Articulated Trucks 700 Series Articulated Trucks	Cat TDTO Cold Weather	SAE 0W-20	−40	−10	−40	14
	Cat TDTO	SAE 30(1)	−20	50	−4	122

(1) Factory fill oil for standard configuration machines.
(2) The minimum allowable oil viscosity at 100° C (212° F)100C is 6.6 cSt (“ASTM D445”).

Backhoe Loaders

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 414 through 450

For hydraulic systems that are not given in Table 10 , refer to the hydraulic system recommendations given in Table 8

Table 10

Backhoe Loaders Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Specification	Oil Viscosity Grade	°C		°F	
			Min	Max	Min	Max
Direct drive transmissions Power shift transmissions	Cat TDTO Cold Weather	SAE 0W-20	−40	35	−40	95
	Cat TDTO	SAE 10W(1)	−20	35	−4	95

(continued)

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(Table 10, contd)

Backhoe Loaders Lubricant Viscosities for Ambient Temperatures						
		SAE 30	25	50	77	122
	Cat TDTO-TMS	Multi-Grade	10	50	50	122
Front Axle Differentials Final Drives for A, B, C, D (Only D with ZF Front axle Series)	**Cat GO (Gear Oil)(2)**	SAE 80W-90(1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	Cat Synthetic GO (2) API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 75W-140	−30	45	−22	113
Front All Wheel Drive Axle Final Drives for D Series (with Cat front axle) E Series F Series	**Cat TDTO Cold Weather**	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	-10	50	14	122
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−20	43	−4	110
Hydraulic Systems for Side Shift Backhoe Loaders	**Cat HYDO Advanced 10**	SAE 10W(1)	−20	50	−4	122
	Cat HYDO Advanced 20(3)	SAE 20	-5	45	23	113
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	0	50	32	122
	Cat Bio HYDO Advanced	“ISO 46” Multi-Grade	−30	45	−22	113
	Cat DEO-ULS Cat DEO	SAE 15W-40	-15	50	5	122
	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104
	Cat ECF-1a, Cat ECF-2, Cat ECF-3	SAE 0W-30	-40	20	−40	68
Brake Reservoir	**Cat HYDO Advanced 10**	SAE 10W	−20	50	−4	122

(1) Factory fill oil for standard configuration machines.
(2) Cat FDAO Syn can be used at a −15° C (5° F) and 50° C (122° F) temperature range
(3) Factory fill for 428B Backhoe Loader.

Backhoe Loader Rear Axles

Do not use Cat MTO or commercial M2C134-D specification oil with the 230-4017 brake disks. Do not use Cat MTO or commercial M2C134-D specification oil in any E Series Backhoe Loader rear axle.

Table 11

Backhoe Loader Rear Axles Lubricant Viscosities for Ambient Temperatures								
Backhoe Loaders	**Brake Disk Part Number**	**197-0017 Additive Volume**	**Oil Type and Performance Requirements for Rear Axle**	**Oil Viscosity Grade**	**°C**		**°F**	
					Min	**Max**	**Min**	**Max**
B, C, D Series	133-7234(1)	1L (1.1qt)(2)	Cat TDTO 30	SAE 30	−25	40	−13	104
	230-4017	150mL (5.1oz)(3)	Cat TDTO 30					
E and F Series except 450E and 450F	238-5291	500 mL (17.0 oz)(4)	Cat TDTO 30					

(continued)

(Table 11, contd)

Backhoe Loader Rear Axles Lubricant Viscosities for Ambient Temperatures								
Backhoe Loaders	Brake Disk Part Number	197-0017 Additive Volume	Oil Type and Performance Requirements for Rear Axle	Oil Viscosity Grade	°C		°F	
					Min	Max	Min	Max
450E 450F	288-7303	200mL (6.8oz)[5]	Cat TDTO 30					

[1] Cat MTO or M2C134-D can be used with 133-7234 brake disks.
[2] The maximum amount of 197-0017 for this brake is 2 L (2.1 qt).
[3] The maximum amount of 197-0017 for this brake is 300 mL (10.2 oz).
[4] The maximum amount of 197-0017 for this brake is 550 mL (18.7 oz).
[5] The maximum amount of 197-0017 for this brake is 250 mL (8.5 oz).

Excavators, Front Shovels, Mass Excavators, Demolition Excavators, and Track Material Handlers

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following model Excavators:

- 301 through 390

This section includes, but not limited to the following model Front Shovels:

- 5090 through 5230 and 385

This section includes, but not limited to the following model Mass Excavators:

- 365 through 385, and 5110 through 5230

This section includes, but not limited to the following model Demolition Excavators:

- 330 through 385

This section includes, but not limited to the following model Track Material Handlers:

- 320 through 385

For hydraulic systems that are not given in Table 12 , refer to the hydraulic system recommendations given in Table 8 .

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Table 12

Excavators, Front Shovels, Mass Excavators, Demolition Excavators, and Track Material Handlers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosity Grade**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Final Drives and Swing Drives for small, medium, and large models **Final Drives** for 307 and 308 mini excavators	**Cat TDTO Cold Weather**	SAE 0W-20	-40	0	-40	32
	Cat TDTO	SAE 10W	−30	0	-22	32
		SAE 30(1)	−30	35	−22	95
		SAE 50	-15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−25	25	−13	77
Track Roller Frame Recoil Spring and Pivot Shaft Bearings	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	−20	25	−4	77
		SAE 50	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	−25	25	−13	77
Track Idlers and Track Rollers	**Cat DEO-ULS Cat DEO**	SAE 15W-40(1)	-10	50	14	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	−35	40	−31	104
	Cat FDAO SYN	Multi-Grade	-30	50	-22	122

(1) Factory fill oil for standard configuration machines.

Forestry

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following model Log Loaders:

- 320 through 345, and 568

This section includes, but not limited to the following model Wheel Feller Bunchers:

- 533 through 573

This section includes, but not limited to the following model Track Feller Bunchers:

- TK711 through TK722
- TK1051 through TK1162
- 521 through 552

This section includes, but not limited to the following model Track Harvester:

- 320 and 501

This section includes, but not limited to the following model Track Skidders:

- 517 through 527

This section includes, but not limited to the following model Wheel Skidders:

- 515 through 545

This section includes, but not limited to the following model Knuckleboom Loaders:

- 519 and 579

This section includes, but not limited to the following model Forwarders:

- 534 through 584

Commercial biodegradable hydraulic oil (HEES) must meet the Cat BF-2 specification.

Cat MTO SAE 15W-40 is the preferred viscosity grade for the hydraulic hood tilt system when the ambient temperature is below −15 °C (+5 °F) and does not go above 40 °C (104 °F).

For hydraulic systems that are not given in Table 13 , refer to the hydraulic system recommendations given in Table 8 .

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Table 13

Forestry Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Power shift transmissions	**Cat TDTO Cold Weather**	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	0	50	32	122
		SAE 50W	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−10	50	14	122
Final Drive and Swing Drive	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(2)	−25	25	−13	77
		SAE 50(3)	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−30	25	−22	72
Final Drive, Swing Drive and Pump Drive for Knuckelboom Loaders **Final Drive and Swing Drive** for Model 501 Track Harvester **Two Speed Transmission and Drive Axles** for Model 584 Forwarders	Cat Synthetic GO	SAE 75W-140	-30	45	-22	113
	Cat GO (Gear Oil)	SAE 80W-90(1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Track Idlers and Track Rollers	**Cat DEO-ULS Cat DEO**	SAE 15W-40(1)	-10	50	14	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	−35	40	−31	104
	Cat FDAO SYN	Multi-Grade	-30	50	-22	122
Variable Pitch Fan	**Cat DEO Cold Weather Cat DEO Cold Weather**	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	−40	50	−40	122
Drive Axles for Wheel Log Skidders (525-545), excluding 525–555D and Wheel Feller Bunchers (533, 573)	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−25	15	−13	59
		SAE 30(1)	−20	43	−4	110
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−30	43	−22	110
Drive Axle/Final Drive for Wheel Log Skidders (525D-555D)	**Cat TDTO**	SAE 10W	−25	0	−13	32
		SAE 30	−20	20	−4	68
		SAE 50(4)	-10	50	14	122
	Cat TDTO-TMS	Multi-Grade	-30	22	-22	72
	Cat TDTO Cold Weather	SAE 0W-20	−40	-10	−40	14
Final Drives for Large Track Skidders (Steel Tracked Machines with Elevated Final Drives except the 561M and 561N)	**Cat FDAO**	SAE 60(1)	−7	50	19	122
	Cat TDTO	SAE 50	−15	32	5	90
		SAE 30	−25	15	−13	59

(continued)

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(Table 13, contd)

Forestry Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
	Cat FDAO SYN	Multi-Grade	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−35	15	−31	59

(1) Factory fill oil for standard configuration machines.
(2) Factory fill oils for swing drives of standard configuration Log Loaders, Model 320 Track Harvester, and Track Feller Bunchers
(3) Factory fill oils for final drives of standard configuration Log Loaders, Model 320 Track Harvester, and Track Feller Bunchers
(4) Factory fill oil for 525–555D standard configuration machines.

Motor Graders

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 12 through 24
- 120 through 163

For the Tandem Drive on the Series M, add 0.015 L (0.015 qt) of 1U-9891 oil additive per 1 L (1 qt) of oil. Do not add oil additive to the 24M Motor Grader or Motor Graders prior to the M series.

For hydraulic systems that are not given in Table 14 , refer to the hydraulic system recommendations given in Table 8 .

Table 14

Motor Graders Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosity Grade	°C		°F	
			Min	Max	Min	Max
Transmission Differential, and Final Drive except 24 and 24M	Cat TDTO Cold Weather	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	0	35	32	95
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	−5	110
Differential, and Final Drive for 24 and 24M	Cat TDTO Cold Weather	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30	0	25	32	77
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	−5	110
Tandem Drive and Wheel Spindle Bearings	Cat TDTO Cold Weather	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	20	−4	68
		SAE 30(1)	−10	40	14	104
		SAE 50	10	50	50	122

(continued)

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(Table 14, contd)

Motor Graders Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosity Grade**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
	Cat TDTO-TMS	Multi-Grade	−15	43	−5	109
All Wheel Drive Gearbox	**Cat TDTO**	SAE 50	−15	50	5	122
Circle Drive except 16H, 24H, 16M, and 24M Series	**Cat Synthetic GO**	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Circle Drive for 16H, 24H, 16M, and 24M Series	**Cat FDAO Syn**	Multi-Grade	−15	50	5	122
	Cat FDAO	SAE 60	-10	50	14	122
	Cat FD-1	SAE 50	-10	32	14	90
Hydraulic Systems for M Series Motor Graders	Cat HYDO Advanced 10	SAE 10W	−15	40	5	104
	Cat HYDO Advanced 30	SAE 30	20	50	68	122
	Cat BIO HYDO Advanced	ISO 46 Multi-Grade	−25	45	−13	113
	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104
	Cat DEO Cold Weather	SAE 0W-40	−25	40	−13	104
	Cat ECF-1a Cat ECF-2 Cat ECF-3	SAE 0W-30	−35	40	−31	104

(1) Factory fill oil for standard configuration machines.

Off Highway Trucks

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 769 through 797
- MT4400D AC, 794 AC,796 AC, and 798 AC

Some Off-Highway Truck torque converters have a common sump with the transmission. These torque converters will use the same fluid recommendations as the transmission. The affected Off-Highway Truck models include 769D, 770, 771D, 772, 773D, 773E, 773F, 775D, 775E, 775F, 793C-F, and all 797 models.

Cat FDAO SYN and Cat FDAO SAE 60 are preferred in Differential, Front Wheel, and Final Drive applications. Cat FDAO SYN is the preferred oil for extremely cold or hot environments, for maximum component life, and for extending the rear axle oil change intervals. If the ambient temperature is below −10°C (14°F) for FDAO SAE 60 or below −35°C (−31°F) for FDAO SYN, warm up the oil prior to operation by performing the procedures in the Operation and Maintenance Manual, “Differential Warm-up and Break-in” prior to operation. Failure to warm up the oil prior to operation can damage the machine.

Note: The minimum acceptable viscosity for commercial alternative oils in most Cat machine hydraulic and hydrostatic transmission systems is 6.6 cSt at 100 °C (212 °F) (“ASTM D445”).

For hydraulic systems that are not given in Table 15 , refer to the hydraulic system recommendations given in Table 8 .

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Table 15

Off Highway Trucks Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Power shift transmissions Except for models listed below and AC models	**Cat TDTO Cold Weather**	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	−10	50	14	122
Power shift transmission in 797 (all models)	**Cat TDTO**	SAE 30(1)	−6	50	−6	122
Power shift transmission for 768C, 768D, 769C, 769D, 770, 771C, 771D, and 772	**Cat TDTO Cold Weather**	SAE 0W-20	−40	22	−40	72
	Cat TDTO	SAE 10W(1)	−20	22	−4	72
		SAE 30	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	—15	43	5	110
Steering Systems except for 785D, 793D, 793F, 795F, 797(all models)	Cat HYDO Advanced 10 Cat TDTO	SAE 10W(1)	−20	40	−4	104
	Cat HYDO Advanced 20 Cat TDTO	SAE 20	-5	45	23	113
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	10	50	50	122
	Cat BIO HYDO Advanced	“ISO 46” Multi-Grade	−30	45	−22	113
	Cat DEO-ULS SYN	SAE 5W-40	−30	40	−22	104
	Cat TDTO Cold Weather	SAE 0W-20	-40	40	-22	104
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat ECF-1-a, Cat ECF-2, Cat ECF-3,	SAE 0W-30	−40	40	−40	104
		SAE 5W-30	−30	40	−22	104
	Cat DEO-ULS Cat DEO	SAE 10W-30	−20	40	−4	104
		SAE 15W-40	−15	50	5	122
	Cat MTO	Cat MTO	−20	40	−4	104
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
Steering system for 785D, 793D, 793F, 795F, 797B, and 797F	Cat HYDO Advanced 30 Cat TDTO	SAE 30(1)	−5	50	23	122
	Cat BIO HYDO Advanced	“ISO 46” Multi-Grade	−30	45	−22	113
	Cat DEO-ULS SYN	SAE 5W-40	−30	40	−22	104
	Cat DEO-ULS Cat DEO	SAE 15W-40	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
	Cat ECF-1-a, Cat ECF-2, Cat ECF-3	SAE 0W-30	−40	27	−40	77
Hoist, Torque Converter, and Brake System	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104
	Cat TDTO	SAE 10W(1)	−20	50	−4	122
Differential, Front Wheels, and Final Drives (2)	Cat FDAO	SAE 60(1)	−10	50	14	122
	Cat FDAO SYN	Multi-Grade	−25	50	-13	122

(continued)

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(Table 15, contd)

Off Highway Trucks Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Hydraulic Systems (includes Brake System, Steering System, Hoist System, and Electric Lubrication (Auto Lube) Pump) for 794 AC, 796 AC, 798 AC,	Cat TDTO	SAE 10W	–20	50	–4	122
	Cat TDTO Cold Weather	SAE 0W-20	–40	40	–40	104

(1) Factory fill oil for standard configuration machines.
(2) Certain truck models up to 777 may use SAE 50 oils under certain operation conditions. These oils include Cat TDTO SAE 50 and commercial Cat TO-4 and Cat FD-1 SAE 50. Refer to the Operation and Maintenance Manual of your truck for details.

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Special Applications for Off Highway Trucks

Table 16

All models for 785, 789, 793, 795, 797 Off-Highway Truck Lubricant Viscosities for Ambient Temperatures						
Compartment or System	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
			Min	Max	Min	Max
Powered Stairway	Cat TDTO	SAE 10W(1)	0	50	32	122
	Cat TDTO Cold Weather	SAE 0W-20	0	50	32	122
	MIL-H-5606A	MIL-H-5606A	-40	40	-40	104

(1) Factory fill oil for standard configuration machines.

Paving

This section includes, but not limited to the following models:

- CS-323 through CS-683
- CP-323 through CP-663
- CB-14 through CB-34
- CB-114 through CB-634
- CC-24 through CC-34
- CD44 through CD54
- CS-44 through CS-76
- CP-44 through CP-76
- AS-2251 through AS-4251
- PS-150 through PS-360
- PF-300
- PM-102 through PM-565
- RM-250 through RM-500
- AP-500 through AP-1055
- BG-225 through BG-2455

For drive axles, Cat TDTO SAE 50 is preferred in most applications, particularly continuous operation. If the ambient temperature is below −15 °C (5 °F), warm up the oil prior to operation by performing the procedures in the OOperation and Maintenance Manual, “Engine and Machine Warm-Up”. Failure to warm up the oil prior to operation can damage the machine.

Do not use API GL-5 or API GL-4 Gear Oils for the Vibratory Compactor Eccentric Weight Housing, Final Drive Planetary Drum, or Vibratory Support. Cat Synthetic Compactor Oil is a premium PAO (Polyalpaolefin) gear and bearing lubricant with no viscosity improvers.

For hydraulic systems that are not given in Table 17 , refer to the hydraulic system recommendations given in Table 8 .

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Table 17

Paving Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Powershift, Manual Transmissions	**Cat TDTO Cold Weather**	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W(1)	−20	10	−4	50
		SAE 30	0	35	32	95
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	5	110
Hydraulic Systems and Hydrostatic Transmissions for Asphalt Compactors Except for Pneumatic Compactors PS-150B s/n: 3XR00621-Up, PS-200B s/n: 5JR00393-Up, PS-360B s/n: 9LS00259-Up, PS-150C, and PS360C	Cat HYDO Advanced 10 Cat TDTO	SAE 10W(1)	−20	40	−4	104
	Cat HYDO Advanced 20 Cat TDTO	(1) SAE 20	-5	45	23	113
	Cat HYDO Advanced 30 Cat TDTO	SAE 30	10	50	50	122
	Cat BIO HYDO Advanced	ISO 46 Multi-Grade	−30	45	−22	113
	Cat TDTO Cold Weather	SAE 0W-20	−40	40	−40	104
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	-30	40	-22	104
	Cat ECF-1-a, Cat ECF-2, Cat ECF-3,	SAE 0W-30	−40	40	−40	104
	Cat DEO Cat DEO-ULS	SAE 10W-30	−20	40	−4	104
		SAE 15W-40	−15	50	5	122
	Cat MTO	SAE 10W-40	−20	40	−4	104
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
Hydraulic Systems and Hydrostatic Transmissions for Pneumatic Compactors PS-150B s/n: 3XR00621-Up, PS-200B s/n: 5JR00393-Up, PS-360B s/n: 9LS00259-Up, PS-150C, and PS360C	**Cat TDTO-TMS**	Multi-Grade	−15	50	5	122
Drive Axles for Medium Compactors (815-826), Vibratory Soil Compactor model numbers with the E or higher suffix	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30	−25	25	−13	77
		SAE 50(1)	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−30	43	−22	110
Drive Axles for Large Compactors	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30	−25	25	−13	77
		SAE 50(1)	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−25	22	−13	72
Axle for the Series 500 Vibratory Soil Compactor models with the D suffix or lower	Cat MTO commercial M2C 134-D	SAE 10W-30	−25	40	−13	104

(continued)

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(Table 17, contd)

Paving Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Vibratory Compactor Eccentric Weight Housing, Final Drive Planetary Drum, and Vibratory Support	Cat Synthetic Compactor Oil	Synthetic ISO 220 (2)	−20	50	−4	122
	Commercial Synthetic Oil	Synthetic ISO 220 (3)	−20	50	−4	122
		Synthetic ISO 68 (4)	−47	21	−53	70

(1) Factory fill oil for standard configuration machines.
(2) This Cat lubricant is a premium PAO (polyalphaolefin) synthetic gear oil with no viscosity improver. This lubricant has an ISO viscosity grade to 220, and a minimum viscosity index of 152.
(3) Commercial oil for this application should have a full synthetic base stock with no viscosity improvers and an ISO viscosity grade to 220, and a minimum viscosity index of 150.
(4) Select a synthetic lubricant with no viscosity improvers and ISO 68 viscosity grade. This lubricant should have a minimum viscosity index of 145 and a minimum pour point of −47° C (−53° F).

Table 18

Special Paving Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Propel Final Drive	**Cat TDTO**	SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−20	43	−4	110
Auger, Conveyor, and Pump Drive Reducers	**Cat GO (Gear Oil)**	SAE 80W-90(1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Track Idlers and Track Rollers	**Cat DEO-ULS Cat DEO**	SAE 15W-40(1)	−10	50	−14	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	-35	40	-31	104
	Cat FDAO SYN	Multi-grade	-30	50	-22	122

(1) Factory fill oil for standard configuration machines.

Pipe Layers

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 561 through 589
- PL61 through PL87

When you operate the machine at temperatures below −18 °C (0 °F), refer to Special Publication, SEBU5898, “Cold Weather Recommendations For All Caterpillar Equipment”. This publication is available from your Cat dealer.

Where recommended for use, Cat FDAO or commercial oil that meets Cat FD-1 are the preferred oil types to maximize gear life and bearing life. Do not use Cat FDAO or Cat FD-1 in compartments that contain clutches and/or brakes. Cat TDTO, Cat TDTO-TMS, or commercial oil that meets Cat TO-4 must be used in any compartment that contains friction material unless you are otherwise specified by Cat.

For the Final Drives in severe usage or in continuous operations, warmup is required. Exercise the final drives for several minutes with the engine at a partial throttle to warmup the oil prior to production operation.

For hydraulic systems that are not given in Table 19 , refer to the hydraulic system recommendations given in Table 8 .

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Table 19

Pipe Layers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Power Shift Transmission and Winch	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	−25	25	−13	77
		SAE 50	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−15	50	5	122
PACCAR Winches	Cat TDTO Cold Weather	SAE 0W-20	-40	10	-40	50
Track Pins	**Cat Synthetic GO**	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90(1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
The Recoil Spring for the Track Roller Frame and the Pivot Shaft Bearings	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	−20	25	−4	77
		SAE 50	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	−25	25	−13	77
Track Idlers and Track Rollers	**Cat DEO-ULS Cat DEO**	SAE 15W-40(1)	−10	50	14	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	−35	40	-31	104
	Cat FDAO SYN	Multi-Grade	-30	50	-22	122
Final Drive(2)	**Cat FDAO**	SAE 60(1)	−7	50	19	122
	Cat TDTO	SAE 50(3)	−15	32	5	90
		SAE 30	−25	15	−13	59
	Cat TDTO-TMS	Multi-Grade	−35	15	−31	59
	Cat FDAO SYN	Multi-Grade	-15	50	5	122

(1) Factory fill oil for standard configuration machines.
(2) These recommendations apply to moderate use or intermittent operation. For severe usage or continuous operation (multiple shifts), follow the severe application recommendations for Track Type Tractors final drives located in Table .
(3) Factory fill for PL61 standard configuration machine.

Skid Steer Loaders

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 216 through 299

For hydraulic systems that are not given in Table 20 , refer to the hydraulic system recommendations given in Table 8 .

PROOF

Table 20

Skid Steer Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Track Idlers and Track Rollers for Compact Track Loaders	**Extreme Pressure Go**	ISO 220(1)	−40	50	−40	122
	Cat Synthetic Go	80W–90	−40	50	−40	122
Final Drive for Multi Terrain Loaders and Compact Track Loaders	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113

(1) Factory fill for standard configuration machines.

Telehandler

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models of Telehandlers:

Note: Refer to the Operation and Maintenance Manual for the most current fluid specifications on Telehandlers.

- TH255 through TH514
- TL642 through TL1255

Table 21

Telehandlers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Hydraulic System, Service Brake System, Transmission, Differentials, and Front Drives for TH220, TH225, TH330, TH360. and Hydraulic systems for TH336, TH337, TH406, TH407, TH414, TH417, TH514, TH306D, TH357D, TH408D, TH3510D, TL642, TL943, TL1055, TL1255 and D series TH642, TH943, TH943, TL1055, TL1255,and Transmission pump for , TH6357D, TH408D, TH603D, TH3510D	**Cat TDTO Cold Weather**	SAE 0W-20	−40	40	−40	104
	Cat TDTO	SAE 10W(1)	−20	40	−4	104
		SAE 30	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−20	50	−4	122
	Cat DEO-ULS Cat DEO	SAE 15W-40	−15	50	5	122
		SAE 10W-30	-20	40	-4	104
	Cat DEO-ULS SYN	SAE 5W-40	−30	40	-22	104
	Cat ECF-1-a Cat ECF-2 Cat ECF-3	SAE 5W-30	-30	40	-22	104
Axles differentials and Wheel Ends for TH336 through TH514, TL642C, and TL943C.	**Cat TDTO**	SAE 30	-20	40	-4	104
Axles differentials and Wheel Ends for TL642, TL943, TL1055, and TL1255. (Except for the C Series) and TH514D	**Cat Synthetic GO**	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90(1)	-23	49	−10	120
		SAE 85W-140	−12	59	10	120
	Cat TDTO-TMS	Multi-Grade	−20	50	−4	122

(continued)

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(Table 21, contd)

Telehandlers Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Axles differentials and Wheel Ends for D seriesTL642, TL943, TL1055, TL1255	**Cat GO(2)**	SAE 80W-90(1)	—20	48	—4	118
	Cat TDTO-TMS(2)	Multi-Grade	−20	48	−4	118
	API GL4 or API GL5 With limited slip additive	SAE 75W-90	-30	45	-22	113
Axles differentials and Wheel Ends for TH306D, TH357D, TH408D, Th3510D	**Cat Synthetic GO(2)**	SAE 75W-140	—30	45	-22	113
	Cat GO (Gear Oil)(2)	SAE 80W-90(1)	—20	50	-4	122
		SAE 85W-140	—10	50	14	122
	API GL5 With limited slip additive	SAE 90	-0	40	32	104
		SAE 80W-90	-20	40	-4	104
		SAE 75W-90	-40	40	-40	104
		SAE 75W	-40	10	-40	50
Transmission and Transfer Case for TL642, TL943, TL1055, and TL1255	Cat MTO	SAE 10W-30(3)	-23	40	-10	104
	Cat TDTO Cold Weather	SAE 0W-20	-40	35	-40	95
	Cat TDTO	SAE 10W(4)	-20	35	-4	95
		SAE 30	25	50	77	122
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	10	50	50	122
	Cat ATF Dexron or Mercon ATF	Multi-Grade Synthetic	-42	-1	-40	14
Transmission and Transfer Case for TH406, TH407, TH514/417, TH336, and TH337	**Cat TDTO Cold Weather**	SAE 0W-20	−40	35	−40	95
Transmission and Transfer Case for D series TL642, TL943, TL1055, TL1255, and TL514D	**Cat TDTO**	SAE 10W(1)	−20	35	−4	95
		SAE 30	25	50	77	122
		SAE 50	10	50	50	122
	Cat ECF-3 Cat ECF-2 Cat ECF-1	SAE 5–30	—30	20	—22	68
		SAE 0–50	-40	20	-40	68
Brake Fluid	**Refer to the Operation and Maintenance Manual for the machine being serviced**					

(1) Factory fill oil for standard configuration machines and dealer service.
(2) Friction modifier 197-0017 is required for D series TL642, TL943, TL1055, TL1255
(3) Factory fill oil for standard configuration machines.
(4) Dealer service fill.

Track Loaders

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- 933 through 973

For hydraulic systems that are not given in Table 22 , refer to the hydraulic system recommendations given in Table 8 .

PROOF

Table 22

Track Loaders Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Final Drive	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	−25	25	−13	77
		SAE 50	−15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−30	25	−22	77
Equalizer Bar End and Pin Joint	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Track Roller Frame Recoil Spring and the Pivot Shaft Bearings	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	−20	25	−4	77
		SAE 50	0	50	32	122
	Cat TDTO-TMS	Multi-Grade	−25	25	−13	77
Track Idlers and Track Rollers	**Cat DEO-ULS Cat DEO**	SAE 15W-40(1)	−10	50	14	122
	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
	Cat DEO-ULS SYN	SAE 5W-40	−35	40	-31	104
	Cat FDAO SYN	Multi-Grade	-30	50	-22	122

(1) Factory fill oil for standard configuration machines.

Track Type Tractors

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- D3 through D11

Where recommended for use, Cat FDAO SYN Cat FDAO or commercial oil that meets Cat FD-1 are the preferred oil types to maximize gear life and bearing life. Do not use Cat FDAO, Cat FDAO SYN, or Cat FD-1 in compartments containing clutches and/or brakes. Cat TDTO, Cat TDTO-TMS, or commercial oil that meets Cat TO-4 oil types must be used in any compartment containing friction material unless specified otherwise by Cat.

For the Final Drives in severe usage or in continuous operations, warmup is required. Exercise the final drives for several minutes with the engine at a partial throttle to warmup the oil prior to production operation.

For hydraulic systems that are not given in Table 23 , refer to the hydraulic system recommendations given in Table 8 .

PROOF

Table 23

Track Type Tractors Lubricant Viscosities for Ambient Temperatures							
Compartment or System	**Application**	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
				Min	**Max**	**Min**	**Max**
Power Shift Transmissions, and Electric Drive Systems	Normal	Cat TDTO Cold Weather	SAE 0W-20	−40	10	−40	50
		Cat TDTO	SAE 10W	−20	10	−4	50
			SAE 30(1)	0	35	32	95
			SAE 50	10	50	50	122
		Cat TDTO-TMS	Multi-grade	−15	43	−5	110
Final Drive for D7E and Elevated Track Type Tractors (Except the D5M, D5N, D6M, and D6N)	Moderate Usage or Intermittent Operation	Cat FDAO	SAE 60(1)	-25	50	-13	122
		Cat TDTO	SAE 50	-25	32	-13	90
			SAE 30	−30	15	−22	59
		Cat TDTO-TMS	Multi-grade	−40	15	−40	59
		Cat FDAO SYN	Multi-grade	−40	50	-40	122
	Severe Usage or Continuous Operation (Multiple Shifts)	Cat FDAO	SAE 60(1)	−35	50	−31	122
		Cat TDTO	SAE 50	−35	14	−31	58
			SAE 30	−40	0	−40	32
		Cat FDAO SYN	Multi-Grade	−40	50	−40	122
Final Drive (Differential Gear Boxes) Oval Track Type Tractors (Including the D5M, D5N, D6M, and D6N) (Except the D7E)	Normal	Cat TDTO Cold Weather	SAE 0W-20	-40	0	-40	32
		Cat TDTO	SAE 10W	-30	0	-22	32
			SAE 30	-25	25	-13	77
			SAE 50(1)	-15	50	5	122

(1) Factory fill oil for standard configuration machines.

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Track Type Tractors Special Applications

Table 24

Special Track Type Tractors Lubricant Viscosities for Ambient Temperatures							
Compartment or System	Application	Oil Type and Performance Requirements	Oil Viscosities	°C		°F	
				Min	Max	Min	Max
End Pin Joints for the Equalizer Bar, Bogie Cartridge Pins, and Track Pins	Normal	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
		Cat GO	SAE 80W-90(1)	−20	40	−4	104
			SAE 85W-140	−10	50	14	122
		API GL-5 Gear Oil	SAE 75W-90	-30	40	-22	104
Winches (hydraulic drive)	Normal	Cat TDTO Cold Weather	SAE 0W-20	−40	0	−40	32
		Cat TDTO	SAE 10W	−20	10	−4	50
			SAE 30(1)	0	43	32	110
		Cat TDTO-TMS	Multi-Grade	−10	35	14	95
Track Roller Frame Recoil Spring Pivot Shaft Bearings	Normal	Cat TDTO Cold Weather	SAE 0W-20	−40	0	−40	32
		Cat TDTO	SAE 10W	−30	0	−22	32
			SAE 30(1)	−20	25	−4	77
			SAE 50	0	50	32	122
		Cat TDTO-TMS	Multi-Grade	−25	25	−13	77
Track Idlers and Track Rollers	Normal	Cat DEO-ULS Cat DEO	SAE 15W-40(1)	-10	50	58	122
		Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
		Cat DEO-ULS SYN	SAE 5W-40	−35	40	−31	104
		Cat FDAO SYN (2)	Multi-Grade	-30	50	-22	122
Variable Pitch Fan	Normal	Cat DEO Cold Weather	SAE 0W-40	−40	40	−40	104
		Cat DEO-ULS SYN	SAE 5W-40	−40	50	−40	122

(1) Factory fill oil for standard configuration machines.
(2) Cat FDAO SYN is required for D11 track idlers.

Underground Mining Equipment

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models:

- AD30 through AD60
- AE40
- R1300 through R3000

For hydraulic systems that are not given in Table 25 , refer to the hydraulic system recommendations given in Table 8 .

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Table 25

Underground Mining Equipment Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Power Shift Transmission	**Cat TDTO Cold Weather**	SAE 0W-20	−40	10	−40	50
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)	0	35	32	95
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	−5	110
Hoist, Torque Converter and Brake System AD45 and AE40	**Cat TDTO Cold Weather**	SAE 0W-20	−40	40	−40	104
	Cat TDTO	SAE 10W	−20	50	−4	122
		SAE 30(1)	-15	50	5	122
Drive Axles AD30 - AD60, and AE40	Cat FD-1	SAE 50	−10	32	14	90
	Cat FDAO	SAE 60(1)	-10	50	14	122
	Cat FDAO SYN	Multi-Grade	-10	50	14	122
Drive Axles Load Haul Dumps R1300 - R3000(2)	**Cat TDTO**	SAE 30	−20	20	−4	68
		SAE 50(1)	−10	43	14	110
	Cat TDTO-TMS	Multi-Grade	−15	22	−5	72

(1) Factory fill oil for standard configuration machines.
(2) R2900 uses SAE 50 fluid if equipped with axle oil cooling.

Wheel Excavators and Wheel Material Handlers

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes, but not limited to the following models of Wheel Excavators and Wheel Material Handlers:

- M312 through M325

For hydraulic systems that are not given in Table 26 , refer to the hydraulic system recommendations given in Table 8 .

Table 26

Wheel Excavators and Wheel Material Handler Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Swing Drives for M325C MH, M325C L MH, M325D MH and M325D L MH	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−30	0	−22	32
		SAE 30(1)	-25	25	-13	77
		SAE 50	-15	50	5	122
	Cat TDTO-TMS	Multi-Grade	−25	25	−13	77

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(Table 26, contd)

Final Drive and Axle for M300A Series, M300C Series with Serial numbers 1 through 2000, M325C MH, M325C L MH, M325D MH and M325D L MH [2]	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90[1]	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	API GL-5 gear oil	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Final Drive and Axle for M300A Series, M300C Series, M300D Series, M325C MH, M325C L MH, M325D MH, and M325D L MH	**Cat MTO**	SAE 10W-30	−25	40	−13	104
	Commercial M2C 134-D					
Transmission	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30	0	35	32	95
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	5	110

[1] Factory fill oil for standard configuration machines.
[2] All models require the use of limited slip additive, Cat 197-0017, Refer to the machine Operation and Maintenance Manual.

Wheel Loaders, Integrated Toolcarriers, Wheeled Dozers, and Compactors

Refer to the “General Information for Lubricants” article for important lubricant information.

This section includes Wheel Loaders and Integrated Toolcarriers, but not limited to the following models:

- 902 through 994 Wheel Loaders
- IT14 through IT62 Integrated Tool Carriers

This section includes Wheeled Dozers and Compactors, but not limited to the following models:

- 814 through 854 Wheeled Dozers
- 816 through 836 Landfill Compactors
- 815 through 825 Soil Compactors

When you are operating the machine in temperatures below −20°C (−4°F), refer to Special Publication, SEBU5898, “Cold Weather Recommendations”. This publication is available from your Cat dealer.

Wheel Loaders, Integrated Toolcarriers, Wheeled Dozers, and Compactors require the use of oil additives 1U-9891 and 185-4771 for axles that have a wet disc and axle shaft speed brakes. Refer to Special Lubricants for more information.

For hydraulic systems that are not given in Table 27 , refer to the hydraulic system recommendations given in Table 8 .

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Table 27

Wheel Loaders, Integrated Toolcarriers, Wheeled Dozers, and Compactors Lubricant Viscosities for Ambient Temperatures						
Compartment or System	**Oil Type and Performance Requirements**	**Oil Viscosities**	**°C**		**°F**	
			Min	**Max**	**Min**	**Max**
Power Shift Transmission, and Electric Drive Systems	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−20	10	−4	50
		SAE 30(1)(2)	0	35	32	95
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−15	43	5	110
	Cat SATO	Multigrade(1) (2)	-20	35	-4	95
Advance Powertrain Transmission (APT)	**Cat SATO**	Multigrade(1)	-30	50	-22	122
	Cat TDTO Cold Weather	SAE 0W-20	−40	20	−40	68
	Cat TDTO-TMS	Multi-Grade	−20	50	-4	122
Final Drive, Differential, and Transfer Drive for 902, 906 and 908 Compact Wheel Loaders	**Cat TDTO**	SAE 30(1)	−20	43	−4	110
Final Drive, Differential, and Transfer Drive for 904 Compact Wheel Loaders	Cat Synthetic GO	SAE 75W-140	−30	45	−22	113
	Cat GO (Gear Oil)	SAE 80W-90(1)	−20	40	−4	104
		SAE 85W-140	−10	50	14	122
	Commercial API GL-5	SAE 75W-90	−30	40	−22	104
		SAE 90	0	40	32	104
Final Drive, Differential, and Transfer Drive for Compact Wheel loaders, 910-914 (Except for 914G2)	**Cat TDTO Cold Weather**	SAE 0W-20	−40	0	−40	32
	Cat TDTO	SAE 10W	−25	15	−13	59
		SAE 30(1)	-20	43	-4	110
		SAE 50	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−30	43	-22	110
Drive Axles for 914 G2 Compact Wheel Loaders	**Cat MTO API GL-4**	SAE 10W-30	-40	40	-40	104
Drive Axles for Small Wheel Loaders (924-938, except for 938K and 938M) Medium Wheel Loaders (950-980) IT12-IT62 Machines Small and Medium Wheel Dozers (814, 824) Medium Compactors (815-826)	**Cat TDTO Cold Weather**	SAE 0W-20	−40	-0	−40	32
	Cat TDTO	SAE 10W	−25	15	−4	59
		SAE 30(1) (3)	−20	43	−4	109
		SAE(4)	10	50	50	122
	Cat TDTO-TMS	Multi-Grade	−25	43	−13	110
Drive Axles for 938K and 938M	**Cat MTO**	SAE 10W-30	-25	40	-13	104
Drive Axles for Large Wheel Loaders (988-993), Large Wheel Dozers (834-854), and Large Compactors (836)	**Cat TDTO Cold Weather**	SAE 0W-20	−40	-10	−40	14
	Cat TDTO	SAE 10W	−25	0	−4	32
		SAE 30	-20	20	-4	68
		SAE 50(1)	-10	50	14	122
	Cat TDTO-TMS	Multi-Grade	−25	22	−13	72

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